

6.7 General Solution of Linear Differential Equations by the Method of Undetermined Coefficients

In the preceding sections we discussed the methods for finding the general solution of the nonhomogeneous linear differential equation $F(D)y = f(x)$. This solution is composed of the sum of the **complementary function** (or, **transient solution**) y_c , and a **particular integral** (or, **steady-state solution**) y_p . In this section we shall discuss another technique called the **method of undetermined coefficients** to obtain a particular integral. Although this method is not applicable in all cases, however, it may be used if the right hand term $f(x)$ is a linear combination of the terms of the types,

- x^n , where n is a non-negative integer,
- e^{mx} , where m is a constant,
- $\sin ax$, $\cos bx$, or $\tan cx$ (where a , b and c are constants), and their reciprocals, or
- A product/linear combination of the above terms.

Note.

By a linear combination of function of functions: $f_1(x)$, $f_2(x)$, $f_3(x)$, ..., $f_n(x)$ is meant an expression, $c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) + \dots + c_n f_n(x)$, where c 's are constants.

Calculation of Particular Integral (or, Steady-State Solution), y_p

The general procedure to calculate a particular integral, y_p , for a non-homogeneous linear differential equation is given as under,

- Assume a trial value of particular integral, y_p , of the form similar to the right hand term $f(x)$ of the given nonhomogeneous linear differential equation, containing undetermined coefficients for each term.
- Obtain requisite derivatives of y_p , assumed above.
- Substitute the values of y_p (assumed above), and all its derivatives, in the given differential equation.
- Thus an identity is obtained in the independent variable.
- Equate coefficients of the like terms to get a set of linear equations.
- Solve these linear equations to obtain the value of the undetermined coefficients.

- g. Substitute the values of these coefficients in the equation for y_p (assumed above) to get the required particular integral.

6.8. Rules for Assuming the Trial Values of y_p

The rules given below may be followed while supposing the trial value of *particular integral* y_p for the non-homogeneous linear differential equation, $F(D) y = f(x)$.

Rule 1. The trial value of y_p should contain terms corresponding to all unlike terms in $f(x)$, as per the table given below.

Rule 2. No two terms should be like each other in y_p .

Rule 3. y_p should not contain any term like a term in the *complementary function* y_c

Rule 4. In case the right hand expression $f(x)$ of the given differential equation is such that the procedure given in the preceding section leads to a trial value of y_p contrary to Rule 3 above, then multiply each term of this trial value by x^k , where k has the least value so that the conditions of Rule 2 and 3 are satisfied.

Scheme for Selection of Terms for Particular Integral

In order to select the trial value of particular integral, y_p , it is necessary to keep in mind the terms contained in the right hand member $f(x)$ of a non-homogeneous linear differential equation $F(D)y = f(x)$. A scheme for the same is given as follows:

TABLE FOR TRIAL VALUES OF PARTICULAR INTEGRAL

Value of $f(x)$	Trial value of y_p
c (constant)	A (constant, to be determined)
x or ax (a being constant)	$A_0 + A_1 x$; (A 's constants)
$a_0 + a_1 x + \dots + a_n x^n$	$A_0 + A_1 x + \dots + A_n x^n$
$a e^{kx}$	$A e^{kx}$
$(a_0 + a_1 x + \dots + a_n x^n) e^{kx}$	$(A_0 + A_1 x + \dots + A_n x^n) e^{kx}$
$\begin{cases} a \sin mx, \text{ or} \\ a \cos mx, \text{ or} \\ a \cos mx + b \sin mx \end{cases}$	$A \cos mx + B \sin mx$

$$\begin{cases} a \sec mx, \text{ or} \\ a \tan mx, \text{ or} \\ a \sec mx + b \tan mx \end{cases} \quad A \sec mx + B \tan mx$$

$$\begin{cases} a \operatorname{cosec} mx, \text{ or} \\ a \cot mx, \text{ or} \\ a \operatorname{cosec} mx + b \cot mx \end{cases} \quad A \operatorname{cosec} mx + B \cot mx$$

$$\begin{cases} ae^{kx}(a_0 + a_1x + \dots + a_nx^n) \cos mx, \text{ or} \\ ae^{kx}(a_0 + a_1x + \dots + a_nx^n) \sin mx, \end{cases} \quad \begin{cases} e^{kx}(A_0 + A_1x + \dots + A_nx^n) \cos mx + \\ e^{kx}(B_0 + B_1x + \dots + B_nx^n) \sin mx, \end{cases}$$

Example 6.8.1

Solve: $y'' + 16y = x^2$.

Solution

Here we find the complementary function as,

$$y_c = A \cos 4x + B \sin 4x. \quad (6.8.1)$$

For a particular integral we assume the trial value of y_p as,

$$y_p = c_0 + c_1x + c_2x^2 \quad (6.8.2)$$

Therefore, $y'_p = c_1x + 2c_2x$; and $y''_p = 2c_2$. Substituting the values of y_p and y''_p in the given differential equation we obtain,

$$2c_2 + 16(c_0 + c_1x + c_2x^2) = x^2.$$

Now equate coefficients of like powers of x in the above equation, to get,

$$16c_2 = 1, 16c_1 = 0, 2c_2 + 16c_0 = 0.$$

Solving these equations simultaneously we obtain,

$$c_2 = 1/16, c_1 = 0, c_0 = -1/128.$$

Therefore, Eqn.(6.8.2) becomes,

$$y_p = \frac{1}{16}x^2 - \frac{1}{128} \quad (6.8.3)$$

Hence, the general solution of the given differential equation is given by equations (6.8.1) and

(6.8.3) as,

$$y = A \cos 4x + B \sin 4x + \frac{1}{16}x^2 - \frac{1}{128}.$$

Example 6.8.2

Solve: $y'' - 3y' + 2y = 2x^3 - 9x^2 = 6x$.

Solution

The complementary function y_c of the given equation is,

$$y_c = c_1 e^x + c_2 e^{2x}. \quad (6.8.4)$$

For a particular integral we suppose the trial value of y_p as,

$$y_p = A_0 + A_1 x + A_2 x^2 + A_3 x^3. \quad (6.8.5)$$

Differentiating this equation twice we get,

$$y'_p = 3A_3 x^2 + 2A_2 x + A_1,$$

$$y''_p = 6A_3 x + 2A_2.$$

Substituting the values of y_p , y'_p , and y''_p in the given equation we get,

$$(2A_2 + 6A_3 x) - 3(A_1 + 2A_2 x + 3A_3 x^2) + 2(A_0 + A_1 x + A_2 x^2 + A_3 x^3) = 2x^3 - 9x^2 + 6x.$$

Equating the coefficients of like powers of x we get,

$$\text{Coefficients of } x^3 : 2A_3 = 2. \quad \text{It gives: } A_3 = 1;$$

$$\text{Coefficients of } x^2 : -9A_3 + 2A_2 = -9. \quad \text{It gives: } A_2 = 0;$$

$$\text{Coefficients of } x : 6A_3 - 6A_2 + 2A_1 = 6. \quad \text{It gives: } A_1 = 0;$$

$$\text{Constants} : 2A_2 - 3A_1 + 2A_0 = 0. \quad \text{It gives: } A_0 = 0.$$

Substituting these values in equation (6.8.5), we obtain,

$$y_p = x^3. \quad (6.8.6)$$

Therefore, Eqns.(6.8.4) and (6.8.6) give the general solution of the given differential equation as,

$$y = c_1 e^x + c_2 e^{2x} + x^3.$$

Example 6.8.3

$$\text{Solve: } (D^2 - 3D + 7)y = 10 e^{2x}.$$

Solution

The complementary function for the given equation is,

$$y_c = e^{3x/2} \left[A \cos \frac{\sqrt{19}}{2} x + B \sin \frac{\sqrt{19}}{2} x \right]. \quad (6.8.7)$$

Keeping in mind the right hand term of the given differential equation, we suppose the value of a particular integral as,

$$y_p = c e^{2x}. \quad (6.8.8)$$

Differentiating this equation twice, we get,

$$y'_p = 2c e^{2x}, \quad y''_p = 4c e^{2x}.$$

Substituting the values of y_p , y'_p , y''_p in the given equation we get,

$$4c e^{2x} - 6c e^{2x} + 7c e^{2x} = 10 e^{2x}.$$

It yield $5c = 10$, or, $c = 2$. Therefore, equation (5. 8. 8) becomes,

$$y_p = 2 e^{2x}. \quad (6.8.9)$$

Hence, the general solution of the given differential equation is given by Eqns (6.8.7) and (6.8.9) as,

$$y = e^{3x/2} \left[A \cos \frac{\sqrt{19}}{2} x + B \sin \frac{\sqrt{19}}{2} x \right] + 2e^{2x}.$$

Example 6.8.4

$$\text{Solve } (D^2 - 3D + 7)y = 10 x e^{2x}.$$

Solution

The complementary function y_c for this equation is the same as given by Eqn (6.8.7). For a particular integral, we consider the trial value as,

$$y_p = e^{2x} (c_0 + c_1 x). \quad (6.8.10)$$

It gives,

$$y'_p = 2 e^{2x} (c_0 + c_1 x) + c_1 e^{2x} = e^{2x} (2c_1 x + 2c_0 + c_1),$$

$$y''_p = e^{2x} (4c_1 x + 4c_0 + 4c_1).$$

Substituting the values of y , y'_p , and y''_p in the given equation we obtain,

$$e^{2x} (5c_1 x + 5c_0 + c_1) = 10 x e^{2x}.$$

Thus,

$$5c_1 = 10, \text{ which gives } c_1 = 2,$$

$$5c_0 + c_1 = 0, \text{ which gives } c_0 = -2/5.$$

Therefore, Eqn.(6.8.10) gives,

$$y_p = e^{2x} \left(-\frac{2}{5} + 2x \right).$$

Hence, the general solution of the given differential equation is,

$$y = y_c + y_p = e^{3x/2} \left\{ A \cos \frac{\sqrt{19}}{2} x + B \sin \frac{\sqrt{19}}{2} x \right\} + e^{2x} \left(2x - \frac{2}{5} \right).$$

Example 6.8.5

$$\text{Solve } y'' - 5y' + 4y = (3x + 2) e^{-2x}.$$

Solution

The complementary function for the given differential equation is,

$$y_c = c_1 e^x + c_2 e^{4x}. \quad (6.8.11)$$

To find a particular integral we suppose,

$$y_p = e^{-2x} (c_0 + c_1 x). \quad (6.8.12)$$

Therefore,

$$y'_p = e^{-2x} (-2c_1 x + c_1 - 2c_0),$$

$$y''_p = e^{-2x} (4c_1 x - 4c_1 + 4c_0).$$

Substituting these values in the given equation and equating the coefficients of the like terms we get,

$$c_1 = 1/6, \quad c_2 = 7/36.$$

Therefore, the general solution is

$$y = c_1 e^x + c_2 e^{4x} + e^{-2x} (x/6 + 7/36).$$

Example 6.8.6

$$\text{Solve } y'' - 3y' + 2y = x^2 e^x.$$

Solution

The complementary function for the given differential equation is,

$$y_c = c_1 e^x + c_2 e^{2x} \quad (6.8.13)$$

For a particular integral we assume,

$$y_p = e^x (c_3 x^2 + c_4 x + c_5) x^k.$$

But e^x is already present in the complementary function y_c , so vide Rule 4 of the preceding section we take $k = 1$ so that no term of y_p is present in y_c . Thus we take modified particular integral as,

$$y_p = e^x (c_3 x^3 + c_4 x^2 + c_5 x). \quad (6.8.14)$$

Differentiating this equation we obtain,

$$\begin{aligned} y'_p &= e^x (3c_3 x^2 + 2c_4 x + c_5) + e^x (c_3 x^3 + c_4 x^2 + c_5 x) \\ &= e^x [c_3 x^3 + (3c_3 + c_4) x^2 + (2c_4 + c_5) x + c_5] \\ y''_p &= e^x [3c_3 x^2 + (6c_3 + 2c_4) x + 2c_4 + c_5] + \\ &\quad e^x [c_3 x^3 + (3c_3 + c_4) x^2 + (2c_4 + c_5) x + c_5] \\ &= e^x [c_3 x^3 + (6c_3 + c_4) x^2 + (6c_3 + 4c_4 + c_5) x + 2c_4 + 2c_5]. \end{aligned}$$

Substituting the values of y_p , y'_p , y''_p in the given equation we get,

$$\begin{aligned} &e^x [c_3 x^3 + (6c_3 + c_4) x^2 + (6c_3 + 4c_4 + c_5) x + 2c_4 + 2c_5] - 3e^x (c_3 x^3 + (3c_3 + c_4) x^2 \\ &+ (2c_4 + c_5) x + c_5) + 2e^x (c_3 x^3 + c_4 x^2 + c_5 x) = x^2 e^x. \end{aligned}$$

Or,

$$\begin{aligned} &e^x [(3c_3 - 3c_3) x^3 + (6c_3 + c_4 - 9c_3 - 3c_4 + 2c_4) x^2 + (6c_3 + 4c_4 + c_5 - 6c_4 - 3c_5 + 2c_5) x \\ &+ 2c_4 + 2c_5 - 3c_5] e^x = e^x x^2 \end{aligned}$$

Or,

$$e^x [-3c_3 x^2 + (6c_3 - 2c_4) x + 2c_4 - c_5] = e^x x^2.$$

Equating the coefficients of the like terms we get,

$$\text{Coefficients of } x^2: \quad -3c_3 = 1. \quad \text{It gives: } c_3 = -1/3;$$

$$\text{Coefficients of } x: \quad 6c_3 - 2c_4 = 0. \quad \text{It gives: } c_4 = -1;$$

$$\text{Constants:} \quad 2c_4 - c_5 = 0. \quad \text{It gives: } c_5 = -2.$$

Therefore, Eqn (6.8.14) becomes,

$$y_p = e^x \left(-\frac{1}{3} x^3 - x^2 - 2x \right). \quad (6.8.15)$$

Hence, Eqns.(6.8.13) and (6.8.15) give the general solution,

$$y = c_1 e^x + c_2 e^{2x} + e^x \left(-\frac{1}{3} x^3 - x^2 - 2x \right).$$

Examples 6.8.7

Solve $y'' + 2y' + 2y = \sin 3x$.

Solution

The complementary function for the given equation is,

$$y_c = e^{-x} (A \cos x + B \sin x).$$

To find a particular integral we assume,

$$y_p = c_1 \cos 3x + c_2 \sin 3x.$$

It gives,

$$y'_p = -3c_1 \sin 3x + 3c_2 \cos 3x,$$

$$y''_p = -9c_1 \cos 3x - 9c_2 \sin 3x.$$

Substituting the values of y_p , y'_p , and y''_p in the given equation and equating the coefficients of the like terms we obtain,

$$c_1 = -6/85, \quad c_2 = -7/85.$$

Thus we get,

$$y_p = \frac{1}{85} (-6 \cos 3x - 7 \sin 3x).$$

Hence, the general solution of the given differential equation is,

$$y = y_c + y_p = e^{-x} (A \cos x + B \sin x) - \frac{1}{85} (6 \cos 3x + 7 \sin 3x).$$

Examples 6.8.8

Solve $y'' + y = x + 2 \cos x + \sin x$.

Solution

Here the complementary function is,

$$y_c = A \cos x + B \sin x.$$

Since some of the terms of y_c are already present in the right hand term of the given equation, so we assume trial value of a particular integral as,

$$y_p = c_0 + c_1 x + (c_2 \cos x + c_3 \sin x) x.$$

We find y'_p and y''_p and then substitute the values of y_p , y'_p , and y''_p in the given equation. On equating the coefficients of the like terms we obtain,

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = -1/2, \quad \text{and} \quad c_3 = 1.$$

Hence we get the general solution as,

$$y = y_c + y_p = A \cos x + B \sin x + x + x \left(-\frac{1}{2} \cos x + \sin x\right).$$

Example 6.8.9

Solve $y''' - 2y'' + y' - 2y = 4x + 5e^{2x} + 20 \cos x$.

Solution

The complementary function is,

$$y_c = c_1 e^{2x} + A \cos x + B \sin x.$$

Keeping in mind the right hand terms of the given differential equation we may assume trial value of y_p as,

$$y_p = c_2 x + c_3 + c_4 e^{2x} + c_5 \cos x + c_6 \sin x.$$

But e^{2x} , $\cos x$ and $\sin x$ are present in y_c . Therefore, we multiply the last three terms of y_p by x . Thus, we assume a revised value of y_p as,

$$y_p = c_2 x + c_3 + c_4 x e^{2x} + c_5 x \cos x + c_6 x \sin x.$$

On successive differentiation and rearranging the terms we obtain,

$$y'_p = c_2 + (c_4 + 2c_4 x) e^{2x} + (c_5 + c_6 x) \cos x + (c_6 - c_5 x) \sin x;$$

$$y''_p = (4c_4 + 4c_4 x) e^{2x} + (2c_6 - c_5 x) \cos x - (2c_5 + c_6 x) \sin x;$$

$$y'''_p = (12c_4 + 8c_4 x) e^{2x} - (3c_5 + c_6 x) \cos x - (3c_6 - c_5 x) \sin x.$$

We substitute these values in the given equation and equate the coefficients of the like terms. Thus we obtain,

$$c_2 = -2, \quad c_3 = -1, \quad c_4 = 1, \quad c_5 = -2, \quad c_6 = -4.$$

Thus the value of y_p becomes,

$$y_p = -2x - 1 + xe^{2x} - 2x \cos x - 4x \sin x.$$

Hence the general solution of the given differential equation is,

$$y = (c_1 + x) e^{2x} + (A - 2x) \cos x + (B - 4x) \sin x - 1 - 2x.$$

EXERCISE-6.3

(Solution of Non-Homogeneous Linear Differential Equations)

Note: Answers are given against the respective questions within square brackets.

Solve the following differential equations,

1. $y'' - 4y = 12$, (Hints : Let $y_p = A$) $[y = c_1 e^{2x} + c_2 e^{-2x} - 3]$
2. $y'' + y' - 2y = 3 - 6x$; $[y = c_1 e^x + c_2 e^{-2x} + 3x]$
3. $y'' + 5y' + 6y = x^2 + x$; $[y = c_1 e^{-2x} + c_2 e^{-3x} + \frac{1}{6}x^2 - \frac{x}{9} + \frac{1}{27}]$
4. $y''' - 3y'' + 3y' - y = 2x^2 - 3x - 17$; $[y = (c_0 + c_1x + c_2x^2)e^x - 2x^2 - 9x + 2]$
5. $y'' + 2y' - 3y = 42e^{4x}$; $[y = c_1 e^{-3x} + c_2 e^x + 2e^{4x}]$
6. $y'' + 5y' + 6y = 2e^{-3x} + 4e^{-2x}$; $[y = c_1 e^{2x} + c_2 e^{3x} - 2xe^{-3x} + 4xe^{-2x}]$
7. $y'' + y = 3 + 6e^x$; $[y = A \cos x + B \sin x + 3 + 3e^x]$
8. $y'' - y' = 6x^5 e^x$; $[y = c + (x^6 - 6x^5 + 30x^4 - 120x^3 + 36x^2 - 720x + c_1)e^x]$
9. $y''' + y' = x \sin x + \cos x$; $[y = c + A \cos x + B \sin x - \frac{5}{4}x \cos x - \frac{1}{4}x^2 \sin x]$
10. $y'' + y' = 6 \sin 2x$; $[y = c_1 + c_2 e^{-x} - \frac{6}{5} \sin 2x - \frac{3}{5} \cos 2x]$
11. $y'' + y = 10e^x \sin x$; $[y = A \cos x + B \sin x + e^x(-4 \cos x + 2 \sin x)]$

Solve the following initial value problems,

12. $y'' - 9y = 18 \cos 3x + 9$; $y(0) = -1$, $y'(0) = 3$; $[y = e^{3x} - 1 - \cos 3x]$
13. $y'' + y' - 6y = 2 - 12x$; $y(0) = 3$, $y'(0) = -7$; $[y = 2x + 3e^{-3x}]$
14. $y'' + y' = e^x$; $y = 3$, $y' = 0$ when $x = \ln 2$; $[y = 1 + 2e^{-x} + \frac{1}{2}e^x]$
15. $y'' + 4y' + 3y = 8xe^x - 6$; while $y(0) = -11/4$, $y'(0) = 1/4$; $[y = xe^x - \frac{3}{4}e^x - 2]$
16. $y''' - 4y' = 6e^{-x} - 3e^x$, while $y = 7$, $y' = 9$, $y'' = 19$, when $x = \ln 2$; $[y = 2e^{2x} + e^x + 2e^{-2x}]$

6.9 Linear Differential Equations with Variable Coefficients

In the preceding sections we have been dealing with linear differential equations with constant coefficients. However, we may come across *linear differential equations with variable coefficients*. It will be of interest to know how such equations could be solved by the use of suitable and often ingenious transformations. It should come as no surprise that some differential equations can be solved by transforming them into linear differential equations with constant coefficients. The particular transformation used may not always be obvious, and one should not develop a sense of inferiority if it is not seen immediately. Such equations include the following types of equations,

a. Cauchy–Euler Equation:

$$a_n x^n D^n y + a_{n-1} x^{n-1} D^{n-1} y + \dots + a_1 x Dy + a_0 y = f(x), \quad (6.9.1)$$

b. Legendre's Equation:

$$(1 - x^2) y'' - 2x y' + n(n-1) y = 0, \quad (6.9.2)$$

c. Gauss's Hypergeometric Equation:

$$x(1-x) y'' + [c - (a+b+1)x] y' - ab y = 0, \quad (6.9.3)$$

d. Bessel's Equation:

$$x^2 y'' + x y' + (x^2 - \nu^2) y = 0, \quad (6.9.4)$$

where the parameter ν is a given number.

Here we shall discuss **Cauchy–Euler equation** (6.9.1) where a 's are constants. This equation can be reduced to a *linear differential equation with constant coefficients* by using the transformation,

$$x = e^t.$$

By taking log-natural of both sides of this transformation we get, $t = \ln x$. Differentiating it with respect to t we get,

$$\frac{dt}{dx} = \frac{1}{x}.$$

Thus,

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt} \frac{1}{x}.$$

Therefore,

$$x \frac{dy}{dx} = \frac{dy}{dt}.$$

Or,

$$x y' = \dot{y}. \quad (6.9.6)$$

Similarly,

$$\begin{aligned}
 y'' &= \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dt} \frac{dt}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dt} \frac{1}{x} \right) \\
 &= \frac{d}{dx} \left(\frac{dy}{dt} \right) \frac{1}{x} - \left(\frac{1}{x^2} \right) \frac{dy}{dt} = \frac{d}{dt} \left(\frac{dy}{dt} \right) \frac{dt}{dx} \frac{1}{x} - \left(\frac{1}{x^2} \right) \frac{dy}{dt} \\
 &= \left[\frac{d^2 y}{dt^2} \right] \frac{1}{x^2} - \frac{1}{x^2} \frac{dy}{dt} = \frac{1}{x^2} \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] = \frac{\ddot{y} - \dot{y}}{x^2}
 \end{aligned}$$

Therefore,

$$x^2 y'' = \ddot{y} - \dot{y} \quad (6.9.7)$$

Proceeding in the same fashion we obtain,

$$\begin{aligned}
 \frac{d^3 y}{dx^3} &= \frac{d}{dx} \left[\frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \right] \\
 &= \frac{d}{dx} (x^{-2}) \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] + \frac{1}{x^2} \frac{d}{dx} \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] \\
 &= \frac{-2}{x^3} \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] + \frac{1}{x^2} \frac{d}{dt} \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] \frac{dt}{dx} \\
 &= \frac{-2}{x^3} \left[\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right] + \frac{1}{x^2} \left[\frac{d^3 y}{dt^3} - \frac{d^2 y}{dt^2} \right] \frac{1}{x} \\
 &= \frac{1}{x^3} \left[-2 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + \frac{d^3 y}{dt^3} - \frac{d^2 y}{dt^2} \right] \\
 &= \frac{1}{x^3} \left[\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right]
 \end{aligned}$$

It gives,

$$x^3 y''' = \ddot{y} - 3\ddot{y} + 2\dot{y}, \quad (6.9.8)$$

and so on.

With a view to simplifying the procedure of solving the Cauchy–Euler equation we make use of the operators,

$$D = \frac{d}{dx}, \text{ and } \Delta = \frac{d}{dt}.$$

Therefore, from Eqns (6.9.6), (6.9.7), (6.9.8), etc. we see that,

$$xD = \Delta$$

$$x^2 D^2 = \Delta^2 - \Delta = \Delta (\Delta - 1).$$

$$x^3 D^3 = \Delta^3 - 3\Delta^2 + 2\Delta = \Delta (\Delta - 1) (\Delta - 2).$$

...

...

$$x^n D^n = \Delta (\Delta - 1) (\Delta - 2) \dots [\Delta - (n - 1)].$$

By making these substitutions in the Cauchy–Euler equation we see that the latter is reduced to a linear differential equation with constant coefficients, having t as the independent variable. Accordingly it can be solved by the methods discussed earlier.

Example 6.9.1

Find the general solution of,

$$x^2 y'' - 2xy' + 2y = 0. \quad (6.9.9)$$

Solution

The equation can be written as,

$$(x^2 D^2 - 2xD + 2)y = 0. \quad (6.9.10)$$

we substitute $x = e^t$. Therefore,

$$xD = \Delta, \text{ and } x^2 D^2 = \Delta(\Delta - 1).$$

Thus, Eqn (6.9.10) becomes,

$$[\Delta(\Delta - 1) - 2\Delta + 2]y = 0. \quad (6.9.11)$$

i.e.,

$$(\Delta^2 - 3\Delta + 2)y = 0.$$

Its auxiliary equation has roots 1, 2. Thus, the general solution of Eqn.(6.9.11) is,

$$y = c_1 e^t + c_2 e^{2t}.$$

But $e^t = x$, and $e^{2t} = x^2$. Therefore, the general solution of the given differential equation is,

$$y = c_1 x + c_2 x^2$$

Example 6.9.2

Solve the differential equation,

$$x^2 y'' + x y' + y = \ln x, (x > 0). \quad (6.9.12)$$

Solution

The given equation can be put as,

$$(x^2 D + xD + 1)y = \ln x \quad (6.9.13)$$

We put $x = e^t$. Therefore, this equation becomes,

$$[\Delta(\Delta-1) + \Delta + 1] y = t,$$

or,

$$[\Delta^2 + 1] y = t \quad (6.9.14)$$

The auxiliary equation of the associated homogeneous equation has the roots $\pm i$. Therefore, the complementary function of equation (6.9.14) is,

$$y_c = A \cos t + B \sin t.$$

A particular integral of equation (6.9.14), its right-hand-side being a polynomial, is given as,

$$\begin{aligned} y_p &= \frac{1}{1 + \Delta^2} [t] = (1 + \Delta^2)^{-1} [t] \\ &= (1 - \Delta^2 + \dots) [t] = t. \end{aligned} \quad (6.9.16)$$

Therefore, the general solution of Eqn (6.9.14) is,

$$y = y_c + y_p = A \cos t + B \sin t + t. \quad (6.6.13)$$

But $e^t = x$, i.e., $t = \ln x$. Thus Eqn (6.6.13) becomes

$$y = A \cos (\ln x) + B \sin (\ln x) + \ln x,$$

which is the general solution of the given differential equation.

EXERCISE-6.4

(Linear Differential Equations with Variable Coefficients)

Note: Answers are given against the respective questions within square brackets.

Find the general solutions to the following differential equations,

1. $x^2 y'' - 4x y' + 6y = 0$; [$y = c_1 x^2 + c_2 x^3$]
2. $4x^2 y'' - 3y = 0$; [$y = c_1 x^{3/2} + c_2 x^{-1/2}$]
3. $x^2 y'' + 6x y' + 6y = x^5$; [$y = c_1 x^{-2} + c_2 x^{-3} + x^5/56$]
4. $x^2 y'' - 2x y' + 2y = x^3$; [$y = c_1 x + c_2 x^2 + x^3/2$]
5. $x^2 y'' - x y' + 2y = \ln x$; [$y = x\{A \cos(\ln x) + B \sin(\ln x) + (\ln x)/2 + 1/2\}$]
6. $x^2 y'' + 5x y' + 4y = x^2 + 16(\ln x)^2$; [$y = c_1/x^2 + (c_2 \ln x)/x^2 + x^2/16 + 4(\ln x)^2 - 8 \ln x + 6$]
7. $x^2 y'' - (2m-1)x y' + (m^2+n^2)y = n^2 x^m \ln x$; [$y = x^m\{A \cos(\ln x^n) + B \sin(\ln x^n) + \ln x\}$]
8. $x^2 y'' - 3x y' + 5y = x^2 \sin(\ln x)$; [$y = x^2\{A \cos(\ln x) + B \sin(\ln x)\} - \frac{x^2}{2} \ln x \cos(\ln x)$]
9. $x^2 y'' - x y' + 4y = \cos(\ln x) + x \sin(\ln x)$;
[$y = x\{A \cos(\sqrt{3} \ln x) + B \sin(\sqrt{3} \ln x)\} + \frac{1}{13}\{3 \cos(\ln x) - 2 \sin(\ln x) + \frac{x}{2} \sin(\ln x)\}$]
10. $(x+1)^2 y'' - 3(x+1)y' + 4y = x^2$;
[$y = (x+1)^2\{c_1 + c_2 \ln(x+1)\} + \frac{1}{2}(x+1)^2\{\ln(x+1)\}^2 - 2x - 7/4$]
11. $(2x+1)^2 y'' - 7(2x+1)y' - 16y = 8(1+2x)^2$;
[$y = (2x+1)^2\{c_1 + c_2 \ln(2x+1) + \{\ln(2x+1)\}^2\}$]
12. $16(x+1)^4 y'''' + 96(x+1)^3 y''' + 104(x+1)^2 y'' + 8(x+1)y' + y = x^2 + 4x + 3$;
[$y = \sqrt{x+1}\{c_1 + c_2 \ln(x+1)\} + \frac{1}{\sqrt{x+1}}\{c_3 + c_4 \ln(x+1)\} + \frac{x^2 + 52x + 51}{225}$]
13. If $x = e^t$, show that, $x y' = \Delta y$, $x^2 y'' = \Delta(\Delta-1)y$, $x^3 y''' = \Delta(\Delta-1)(\Delta-2)y$, where
 $\Delta = d/dt$. Hence solve, $x^3 y''' + 3x^2 y'' = 1 + x$; [$y = c_1 + c_2/x + c_3 x - \ln x + x(\ln x)/2$]
14. $x^2 y''' + 3x y'' - y' = x$; [$y = c_0 + A \cos(\ln x) + B \sin(\ln x) + x^2/10$]
15. $x^3 y''' + 2x^2 y'' + 2y = 10x + 10/x$
[$y = x\{A \cos(\ln x) + B \sin(\ln x) + 5\}x^{-1}\{C + 2 \ln x\}$]
16. $x^4 y'''' + 6x^3 y''' + 9x^2 y'' + 3x y' + y = (1 + \ln x)^2$;
[$y = (c_1 + c_2 \ln x) \sin(\ln x) + (c_3 + c_4 \ln x) \cos(\ln x) + (\ln x)^2 + 2 \ln x - 3$]

$$17. \quad x^4 y''' + 2x^3 y'' - x^2 y' + x y = 1; \quad [y = x(c_1 + c_2 \ln x) + c_3/x + (\ln x)/4x]$$

Solve the initial value problems,

$$18. \quad x^2 y'' + x y' - y = 0; \quad y(1) = 2, \quad y'(1) = 0; \quad [y = x + 1/x]$$

$$19. \quad x^2 y'' - x y' + y = 8x^3, \quad y(1) = 4, \quad y'(1) = 8; \quad [y = 2(x + x^3)]$$

$$20. \quad x^2 y'' + 2x y' - 6y = 10x^2, \quad y(1) = 1, \quad y'(1) = -6; \quad [y = -x^2 + 2/x^3 + 2x^2 \ln x]$$

$$21. \quad x^3 y''' + 2x^2 y'' - x y' + y = 24x^2 + 32/x^3, \quad y(1) = 0, \quad y'(1) = 4, \quad y''(1) = 0; \\ [y = 8x^2 - 10x + 6x \ln x + 3/x - 1/x^3]$$

$$22. \quad x^2 y'' - 6y = \ln x; \quad y(1) = 1/6, \quad y'(1) = -1/6; \quad [y = \frac{1}{6} (1/2x^2 + x^3/3 - \ln x + 1/6)]$$

6.10 The Method of Variation of Parameters

The method of *variation of parameters* (that may also be called the method of *variation of constants*) was devised by the great Italian-French mathematician, **Joseph Louis Lagrange (1736 -1813)**. Lagrange and the Swiss mathematician **Leonhard Euler (1707-1783)** were the two great mathematician of the seventeenth century. **Euler** discovered the general solution to the homogeneous differential equation,

$$ay'' + by' + cy = 0,$$

while **Lagrange** found a particular solution to the non-homogeneous differential equation,

$$ay'' + by' + cy = f(x).$$

This is a powerful method to find a particular solution to a nonhomogeneous linear differential equation once its complementary function is known. The main advantage of this method is that it is a general method, and can be applied to any differential equation. We shall use this method to derive a formula for a particular solution of an arbitrary second order nonhomogeneous linear differential equation. This method applies even when the coefficients of the differential equation are variables, provided we know a fundamental solution set for the associated homogeneous differential equation. Moreover, this method also has the advantage that no special case arises even if the nonhomogeneous term is included in the complementary function.

In order to develop the method we start from the first order linear differential equation,

$$y' + P(x)y = Q(x), \quad (6.10.1)$$

whose solution is given as, $y = e^{-\int P dx} [\int e^{\int P dx} Q(x) dx + c]$. It can be written as,

$$y = e^{-\int P dx} \int e^{\int P dx} Q(x) dx + c e^{-\int P dx}. \quad (6.10.2)$$

We see that the above solution has the form, $y = y_p + y_c$, where,

$$y_c = c e^{-\int P dx}, \quad (6.10.3)$$

is a solution of the associated homogeneous differential equation,

$$y' + P(x)y = 0, \quad (6.10.4)$$

and,

$$y_p = e^{-\int P dx} \int e^{\int P dx} Q(x) dx, \quad (6.10.5)$$

is a particular solution of Eqn.(6.10.1).

In order to derive/develop the **method of variation of parameters**, we rederive the solution to the above differential equation.

Suppose y_1 is a known solution of the homogeneous differential equation (6.10.4). Then,

$$y_1' + P(x)y_1 = 0. \quad (6.10.6)$$

It may be seen that $y_1 = e^{-\int p \, dx}$ is a solution of this differential equation. This equation being linear, its general solution is given as,

$$y = c_1 y_1(x) = c_1 e^{-\int p \, dx}. \quad (6.10.7)$$

The method of variation of parameters consists of finding a function $u_1(x)$ such that,

$$y_p = u_1(x) y_1(x), \quad (6.10.8)$$

is a particular solution of the differential equation (6.10.1). In other words, we replace the parameter c_1 , in Eqn.(6.10.7), by a variable $u_1(x)$. Accordingly, substituting $y_p = u_1 y_1$, in Eqn. (6.10.1), we obtain,

$$\frac{d}{dx} [u_1 y_1] + P(x) [u_1 y_1] = Q(x);$$

i.e.,

$$u_1 y_1' + u_1' y_1 + P(x) u_1 y_1 = Q(x);$$

or,

$$u_1 [y_1' + P(x) y_1] + y_1 u_1' = Q(x).$$

But, $y_1' + P(x) y_1 = 0$, by Eqn.(6.10.6). Thus, the above equation becomes, $y_1 u_1' = Q(x)$, i.e.,

$$y_1 \frac{du_1}{dx} = Q(x). \quad (6.10.9)$$

Therefore,

$$\int du_1 = \int \frac{Q(x)}{y_1(x)} dx;$$

i.e.,

$$u_1 = \int \frac{Q(x)}{y_1(x)} dx.$$

From this equation and Eqn.(6.10.8) it follows that,

$$y_p = u_1 y_1 = y_1 \int \frac{Q(x)}{y_1(x)} dx. \quad (6.10.10)$$

From the value of $y_1(x) = e^{-\int p \, dx}$ (see Eqn.(6.10.7), it may be seen that the above equation is same as Eqn.(6.10.5).

Next, we extend the above mentioned method for first order linear differential equations to the second order linear differential equations. Accordingly, we consider,

$$a(x) y'' + b(x) y' + c(x) y = f(x), \quad (6.10.11)$$

where, $a(x)$, $b(x)$, and $c(x)$ need not necessarily be constants. Dividing this equation throughout by $a(x)$, it can be written in the standard form as,

$$y'' + P(x) y' + Q(x) y = R(x). \quad (6.10.12)$$

Here we assume that the functions $P(x) = \frac{b(x)}{a(x)}$, $Q(x) = \frac{c(x)}{a(x)}$, and $R(x) = \frac{f(x)}{a(x)}$, are continuous on some interval I. The homogeneous differential equation associated with Eqn.(6.10.12) is,

$$y'' + P(x) y' + Q(x) y = 0. \quad (6.10.13)$$

Suppose $y_1(x)$, and $y_2(x)$ are two known linearly independent solutions of this equation. Thus the complementary function (or, transient solution) is,

$$y_c = c_1 y_1 + c_2 y_2. \quad (6.10.14)$$

Like in the case of first order linear differential equations discussed above, the idea underlying the method of variation of parameters is to replace the constants c_1 and c_2 by the unknown functions (*variable parameters*), $u_1(x)$ and $u_2(x)$ respectively in Eqn.(6.10.14), and then to seek a particular integral of the non-homogeneous linear differential equation (6.10.12), of the form,

$$y_p = u_1(x) y_1(x) + u_2(x) y_2(x). \quad (6.10.15)$$

In order to determine the values of $u_1(x)$ and $u_2(x)$, we require two equations. They are obtained as under:

- a. Differentiating Eqn.(6.10.15) w.r.t. x we get,

$$\begin{aligned} y_p' &= u_1 y_1' + u_1' y_1 + u_2 y_2' + u_2' y_2. \\ &= (u_1 y_1' + u_2 y_2') + (u_1' y_1 + u_2' y_2). \end{aligned}$$

We require $u_1(x)$ and $u_2(x)$ to be such that the last two terms vanish, i.e.,

$$u_1' y_1 + u_2' y_2 = 0. \quad (6.10.16)$$

This is the **first condition** imposed on $u_1(x)$ and $u_2(x)$. Thus the above equation reduces to,

$$y_p' = u_1 y_1' + u_2 y_2'. \quad (6.10.17)$$

- b. Again differentiating Eqn.(6.10.17) we get,

$$y_p'' = u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2' y_2'.$$

Accordingly substituting the values of y_p , y_p' , and y_p'' into Eqn.(6.10.12), we obtain,

$$(u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2' y_2') + P (u_1 y_1' + u_2 y_2') + Q(u_1 y_1 + u_2 y_2) = R(x).$$

i.e.,

$$u_1 [y_1'' + P y_1' + Q y_1] + u_2 [y_2'' + P y_2' + Q y_2] + (u_1' y_1' + u_2' y_2') = R(x).$$

But by Eqn.(6.10.13), $y_1'' + P y_1' + Q y_1 = 0 = y_2'' + P y_2' + Q y_2$. Therefore, the above equation reduces to,

$$u_1' y_1' + u_2' y_2' = R(x). \quad (6.10.18)$$

This is the **second condition** on $u_1(x)$ and $u_2(x)$.

Thus Eqns. (6.10.16) and (6.10.18) constitute a linear system of equations from which we can determine the values of u_1' and u_2' . By Cramer's rule, the solution of these equations is given as,

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ R(x) & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} = \frac{-y_2 R(x)}{W}, \quad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & R(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} = \frac{y_1 R(x)}{W},$$

(6.10.19)

where the determinant $W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is the **Wronskian of y_1 and y_2** . From these equations we can find u_1 and u_2 by integration, which on substitution into Eqn.(6.10.15) yields y_p . Accordingly the general solution is given as, $y = y_c + y_p$.

The method of *Variation of Parameters* is summarized as follows:

Summary of the Method of Variation of Parameters for Second Order Ordinary Differential Equations

The method of *variation of parameters* is summarized as under:

Step 1. Write the given differential equation in the standard form,

$$y'' + P(x) y' + Q(x) = R(x).$$

Step 2. Find *two linearly independent* solutions $y_1(x)$, and $y_2(x)$ of the associated homogeneous differential equation.

Step 3. Assume that the particular integral of the above equation is $y_p = u_1 y_1 + u_2 y_2$. Accordingly set $y_p' = u_1 y_1' + u_2 y_2'$, by requiring $u_1' y_1 + u_2' y_2 = 0$.

Step 4. Obtain $y_p'' = u_1 y_1'' + u_1' y_1' + u_2 y_2'' + u_2' y_2'$.

Step 5. Substitute the values of y_p , y_p' , and y_p'' in the equation at Step 1, and after simplification obtain, $u_1' y_1' + u_2' y_2' = R(x)$.

Step 6. Solve the equations $u_1' y_1 + u_2' y_2 = 0$, and $u_1' y_1' + u_2' y_2' = R(x)$, simultaneously to get the values of u_1' and u_2' as, $u_1' = \frac{-y_2 R(x)}{W(x)}$, and $u_2' = \frac{y_1 R(x)}{W(x)}$. Integrate these to

find the values of u_1 and u_2 . Here, $W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is the Wronskian of y_1 and y_2 .

Step 7. Accordingly, obtain a ***particular integral*** of the given differential equation as,

$$y_p = y_1(x) \int \frac{-y_2 R(x)}{W(x)} dx + y_2(x) \int \frac{y_1 R(x)}{W(x)} dx$$

Example 6.10.1

$$\text{Solve: } y'' - 4y' + 4y = (x + 1) e^{2x} \quad (6.10.20)$$

Solution

The homogeneous differential equation associated with the given equation is,

$$y'' - 4y' + 4y = 0 \quad (6.10.21)$$

The auxiliary equation for this equation has the roots: 2, 2 (repeated roots). Thus the complimentary function for Eqn.(6.10.20) is,

$$y_c = c_1 e^{2x} + c_2 x e^{2x}. \quad (6.10.22)$$

Therefore, $y_1 = e^{2x}$ and $y_2 = x e^{2x}$ are the two solutions of Eqn.(6.10.21). Accordingly, the Wronskian of y_1 , and y_2 is,

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = 2x e^{4x} + e^{4x} - 2x e^{4x} = e^{4x} \neq 0.$$

It shows that the two solutions y_1 and y_2 are linearly independent. Thus Formulae (6.10.19) give,

$$u_1' = \frac{-y_2 R(x)}{W} = \frac{-x e^{2x} (x+1) e^{2x}}{e^{4x}} = -x^2 - x,$$

which gives on integration,

$$u_1 = \frac{-x^3}{3} - \frac{x^2}{2},$$

and,

$$u_2' = \frac{y_1 R(x)}{W} = \frac{e^{2x} (x+1) e^{2x}}{e^{4x}} = x + 1,$$

which gives on integration,

$$u_2 = \frac{x^2}{2} + x.$$

Therefore, Eqn.(6.10.15), i.e., $y_p = u_1 y_1 + u_2 y_2$, gives,

$$\begin{aligned} y_p &= \left(\frac{-x^3}{3} - \frac{x^2}{2} \right) e^{2x} + \left(\frac{x^2}{2} + x \right) x e^{2x} \\ &= e^{2x} \left[-\frac{x^3}{3} - \frac{x^2}{2} + \frac{x^3}{2} + x^2 \right] = e^{2x} \left[\frac{x^3}{6} + \frac{x^2}{2} \right]. \end{aligned}$$

Hence the general solution of the given differential equation is,

$$\begin{aligned} y &= y_c + y_p \\ &= c_1 e^{2x} + c_2 x e^{2x} + e^{2x} \left[\frac{x^3}{6} + \frac{x^2}{2} \right]. \end{aligned} \quad (6.10.23)$$

Example 6.10.2

Solve: $x^2 y'' - 3xy' + 4y = \ln x$, where $x > 0$. (6.10.24)

Solution

This is the Cauchy-Euler equation. Setting $x = e^t$, $xD = \Delta$, $x^2 D^2 = \Delta(\Delta-1)$, where, $D = \frac{d}{dx}$, and $\Delta = \frac{d}{dt}$, as discussed earlier in this chapter (see sec. 6.9), the associated homogeneous differential equation becomes,

$$[\Delta(\Delta-1) - 3\Delta + 4] y = 0.$$

Its auxiliary equation, $m(m-1) - 3m + 4 = 0$, gives, $m = 2, 2$. Therefore, the complementary function for the given differential equation is,

$$y_c = (c_0 + c_1 t) e^{2t} = (c_0 + c_1 \ln x) x^2.$$

Thus we get two linearly independent solutions as,

$$y_1(x) = x^2, \text{ and } y_2(x) = x^2 \ln x.$$

The Wronskian of these solutions is,

$$W(x) = \begin{vmatrix} x^2 & x^2 \ln x \\ 2x & x + 2x \ln x \end{vmatrix} = x^3.$$

Before finding a particular integral of the given differential equation by the method of *variation of parameters*, we divide it throughout by x^2 to write it in the standard form as,

$$y'' - (3/x)y' + (4/x)y = (\ln x)/x^2, \text{ where } x > 0.$$

Thus $R(x) = (\ln x)/x^2$. Accordingly we obtain a particular integral as,

$$\begin{aligned} y_p &= y_1(x) \int \frac{-y_2 R(x)}{W(x)} dx + y_2(x) \int \frac{y_1 R(x)}{W(x)} dx \\ &= x^2 \int \frac{-(\ln x)^2}{x^3} dx + x^2 \ln x \int \frac{\ln x}{x^3} dx. \end{aligned}$$

Integration by parts gives,

$$\int \frac{-(\ln x)^2}{x^3} dx = \frac{1}{2} \frac{(\ln x)^2}{x^2} + \frac{1}{2} \frac{\ln x}{x^2} + \frac{1}{4} \frac{1}{x^2}$$

and,

$$\int \frac{\ln x}{x^3} dx = -\frac{1}{2} \frac{\ln x}{x^2} - \frac{1}{4} \frac{1}{x^2}.$$

Therefore, substituting these values in the above expression for y_p we get,

$$\begin{aligned} y_p &= x^2 \left[\frac{1}{2} \frac{(\ln x)^2}{x^2} + \frac{1}{2} \frac{\ln x}{x^2} + \frac{1}{4} \frac{1}{x^2} \right] + x^2 \ln x \left[-\frac{1}{2} \frac{\ln x}{x^2} - \frac{1}{4} \frac{1}{x^2} \right] \\ &= \frac{1}{4} + \frac{1}{4} \ln x, \text{ where } x > 0. \end{aligned}$$

Accordingly, the general solution of the given differential equation is given as,

$$y = y_c + y_p = y_c = (c_0 + c_1 \ln x) x^2 + \frac{1}{4} + \frac{1}{4} \ln x, \text{ where } x > 0. \quad (6.10.25)$$

Example 6.10.3

Solve: $y'' + y = \operatorname{cosec} x$, where $x \neq n\pi$, for $n = 1, 2, 3, \dots$. (6.10.26)

Solution

It may be seen that the above differential equation cannot be solved by the method of *undetermined coefficients*. The associated homogeneous differential equation has the auxiliary equation, $m^2 + 1 = 0$, its roots being $m = \pm i$. Therefore, the complementary function is,

$$y_c = A \cos x + B \sin x.$$

Thus the two linearly independent solutions are:

$$y_1 = \cos x, \quad y_2 = \sin x.$$

Therefore, the Wronskian is,

$$W(x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1.$$

Accordingly a particular integral of the given differential equation is,

$$\begin{aligned} y_p &= y_1(x) \int \frac{-y_2 R(x)}{W(x)} dx + y_2(x) \int \frac{y_1 R(x)}{W(x)} dx \\ &= \cos x \int -1 dx + \sin x \int \cot x dx = -x \cos x + \sin x (\ln |\sin x|). \end{aligned}$$

Therefore, the general solution is,

$$y = y_c + y_p = A \cos x + B \sin x - x \cos x + \sin x (\ln |\sin x|). \quad (6.10.27)$$

Here, $x \neq n\pi$, for $n = 1, 2, 3, \dots$, since $\ln |\sin n\pi| = \infty$.

Example 6.10.4

Solve

$$y'' + 2y' + y = x e^{-x}. \quad (6.10.28)$$

Solution

The homogeneous equation associated with the given differential equation has the auxiliary equation $m^2 + 2m + 1 = 0$, whose roots are $m = -1, -1$. Therefore, the complementary function is,

$$y_c = (c_0 + c_1 x) e^{-x} = c_0 e^{-x} + c_1 x e^{-x}.$$

Thus the two linearly independent solutions are, $y_1 = e^{-x}$, and $y_2 = x e^{-x}$, which give the Wronskian as,

$$W(x) = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x} - xe^{-x} \end{vmatrix} = e^{-2x}.$$

Therefore, a particular integral is given as,

$$\begin{aligned} y_p &= y_1(x) \int \frac{-y_2 R(x)}{W(x)} dx + y_2(x) \int \frac{y_1 R(x)}{W(x)} dx \\ &= e^{-x} \int -x^2 dx + x e^{-x} \int x dx = e^{-x} [-x^3/3 + x^3/2] = (x^3 e^{-x})/6. \end{aligned}$$

Thus the general solution of the given differential equations is given as,

$$\begin{aligned} y &= y_c + y_p = (c_0 + c_1 x) e^{-x} + (x^3 e^{-x})/6. \\ &= (c_0 + c_1 x + x^3/6) e^{-x}. \end{aligned} \quad (6.10.29)$$

The Method of Variation of Parameters for Third Order Ordinary Differential Equations

On the analogy of this method for the second order differential equations, it can be extended to third order differential equations as under:

Step 1. Write the given differential equation in the standard form,
 $y''' + P(x) y'' + Q(x) y' + R(x) y = f(x).$ (6.10.30)

Step 2. Find *three linearly independent* solutions $y_1(x)$, $y_2(x)$ and $y_3(x)$ of the associated homogeneous differential equation,

$$y''' + P(x) y'' + Q(x) y' + R(x) y = 0. \quad (6.10.31)$$

Step 3. Assume that the particular integral of the above equation is

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3.$$

Find y_p' , y_p'' , and y_p''' . Accordingly set,

$$y_p' = u_1 y_1' + u_2 y_2' + u_3 y_3', \text{ by imposing condition: } u_1' y_1 + u_2' y_2 + u_3' y_3 = 0,$$

$$y_p'' = u_1 y_1'' + u_2 y_2'' + u_3 y_3'', \text{ by imposing condition: } u_1' y_1' + u_2' y_2' + u_3' y_3' = 0.$$

On differentiating y_p'' , we have,

$$y_p''' = u_1 y_1''' + u_2 y_2''' + u_3 y_3''' + u_1' y_1'' + u_2' y_2'' + u_3' y_3''.$$

Step 4. Substitute the values of y_p , y_p' , y_p'' and y_p''' in Eqn.(6.10.30) at Step 1. Since y_1 , y_2 , and y_3 satisfy the homogeneous differential equation (6.10.31) at Step 2, so we obtain,

$$u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = f(x).$$

This is the ***third condition***.

Step 5. Solve the following equations (obtained at steps 3 and 4, above), simultaneously for u_1' , u_2' and u_3' :

$$u_1' y_1 + u_2' y_2 + u_3' y_3 = 0,$$

$$u_1' y_1' + u_2' y_2' + u_3' y_3' = 0$$

$$u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = f(x).$$

Thus, by Cramer's rule the above equations yield,

$$u_1' = \frac{(y_2 y_3' - y_2' y_3) f(x)}{W(x)}, u_2' = \frac{-(y_1 y_3' - y_1' y_3) f(x)}{W(x)}, u_3' = \frac{(y_1 y_2' - y_1' y_2) f(x)}{W(x)}.$$

where, $W(x) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$ is the Wronskian. Integrate these equations to find u_1 , u_2 , and u_3 .

Step 6. Accordingly, obtain a *particular integral* of the given differential equation as,

$$\begin{aligned} y_p = & y_1(x) \int \frac{(y_2 y_3' - y_2' y_3) f(x)}{W(x)} dx + y_2(x) \int \frac{-(y_1 y_3' - y_1' y_3) f(x)}{W(x)} dx \\ & + y_3(x) \int \frac{(y_1 y_2' - y_1' y_2) f(x)}{W(x)} dx. \end{aligned} \quad (6.10.32)$$

Example 6.10.5

$$\text{Solve } y''' - y'' - y' + y = e^x. \quad (6.10.33)$$

Solution

The homogeneous differential equation associated with the given equation is,

$$y''' - y'' - y' + y = 0. \quad (6.10.34)$$

Its solution, which is the *complementary function* of the given differential equation, is

$$y_c = (c_0 + c_1 x) e^x + c_2 e^{-x} = c_0 e^x + c_1 x e^x + c_2 e^{-x}. \quad (6.10.35)$$

Therefore, there are three linearly independent solutions: $y_1 = e^x$, $y_2 = x e^x$, $y_3 = e^{-x}$. Accordingly we assume a particular integral of the given differential equation (6.10.33) as,

$$y_p = u_1 e^x + u_2 x e^x + u_3 e^{-x}. \quad (6.10.36)$$

On differentiation, this equation gives,

$$y_p' = (u_1 e^x + u_2 x e^x + u_3 e^{-x}) + (u_1' e^x + u_2' x e^x + u_3' e^{-x}).$$

Here we impose the *first condition*,

$$(u_1' e^x + u_2' x e^x + u_3' e^{-x}) = 0, \quad (6.10.37)$$

so that the above equation becomes,

$$y_p' = u_1 e^x + u_2 e^x + u_2 x e^x - u_3 e^{-x}. \quad (6.10.38)$$

On differentiating it we obtain,

$$y_p'' = (u_1 e^x + 2u_2 e^x + u_2 x e^x + u_3 e^{-x}) + (u_1' e^x + u_2' e^x + u_2' x e^x - u_3' e^{-x}).$$

Here, we impose the **second condition**,

$$u_1' e^x + u_2' (1 + x) e^x - u_3' e^{-x} = 0. \quad (6.10.39)$$

Thus the above equation reduces to,

$$y_p'' = u_1 e^x + 2u_2 e^x + u_2 x e^x + u_3 e^{-x}. \quad (6.10.40)$$

Differentiating it we obtain,

$$y_p''' = (u_1 e^x + 3u_2 e^x + u_2 x e^x - u_3 e^{-x}) + (u_1' e^x + 2u_2' e^x + u_2' x e^x + u_3' e^{-x}). \quad (6.10.41)$$

We substitute the values of y_p , y_p' , y_p'' and y_p''' , from Eqns.(6.10.36), (6.10.38), (6.10.40) and (6.10.41) into the given differential equation (6.10.33). On simplification it gives,

$$u_1' e^x + u_2' (2 + x) e^x + u_3' e^{-x} = e^x. \quad (6.10.42)$$

It is the **third condition**. We solve the system of equations (6.10.37), (6.10.39) and (6.10.42) for u_1' , u_2' , and u_3' , as,

$$\begin{bmatrix} e^x & x e^x & e^{-x} \\ e^x & (1+x) e^x & -e^{-x} \\ e^x & (2+x) e^x & e^{-x} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ e^x \end{bmatrix}.$$

Here, the Wronskian, $W(x) = e^x \begin{vmatrix} 1 & x & 1 \\ 1 & 1+x & -1 \\ 1 & 2+x & 1 \end{vmatrix} = 4 e^x$. Therefore, by Cramer's rule,

$$u_1' = \frac{\begin{vmatrix} 0 & x e^x & e^{-x} \\ 0 & (1+x) e^x & -e^{-x} \\ e^x & (2+x) e^x & e^{-x} \end{vmatrix}}{W(x)} = \frac{e^x (-1-2x)}{4 e^x} = \frac{-1-2x}{4},$$

$$u_2' = \frac{\begin{vmatrix} e^x & 0 & e^{-x} \\ e^x & 0 & -e^{-x} \\ e^x & e^x & e^{-x} \end{vmatrix}}{W(x)} = \frac{2e^x}{4 e^x} = \frac{1}{2},$$

$$u_3' = \frac{\begin{vmatrix} e^x & x e^x & 0 \\ e^x & (1+x) e^x & 0 \\ e^x & (2+x) e^x & e^x \end{vmatrix}}{W(x)} = \frac{e^{3x}(1+x-x)}{4e^x} = \frac{e^{2x}}{4}.$$

Integrating the above equations we obtain,

$$u_1 = -\frac{x}{4} - \frac{x^2}{4}, u_2 = \frac{x}{2}, \text{ and } u_3 = \frac{e^{2x}}{8}.$$

Substituting the values of u_1 , u_2 , and u_3 in Eqn.(6.10.36) we obtain,

$$\begin{aligned} y_p &= \left(-\frac{x}{4} - \frac{x^2}{4}\right) e^x + \left(\frac{x}{2}\right) x e^x + \left(\frac{e^{2x}}{8}\right) e^{-x}, \\ &= \frac{1}{8} e^x - \frac{1}{4} x e^x + \frac{x^2}{4} e^x. \end{aligned} \quad (6.10.43)$$

Therefore, Eqns.(6.10.35) and (6.10.43) give the general solution of the given differential equation as,

$$\begin{aligned} y &= y_c + y_p = (c_0 e^x + c_1 x e^x + c_2 e^{-x}) + \left(\frac{1}{8} e^x - \frac{1}{4} x e^x + \frac{x^2}{4} e^x\right). \\ &= (c_0 + \frac{1}{8}) e^x + (c_1 - \frac{1}{4}) x e^x + c_2 e^{-x} + \frac{x^2}{4} e^x. \end{aligned}$$

Or,

$$y = (A + Bx) e^x + C e^{-x} + \frac{x^2}{4} e^x,$$

where A, B and C are constants. This is the solution of the given differential equation, obtained by the method of variation of parameters.

6.11 Green's Function

Green's functions are named after an English mathematician **George Green (1793-1841)**. He was a self taught mathematician who made many significant contributions in the fields of electricity, magnetism, fluid mechanics and partial differential equations. He introduced the functions now named after his name, the **Green's functions**, which help us in solving the boundary value problems. Moreover, he developed the integral transformation theorems of which Green's theorem in the plane is a particular case. These results became widely known after Green's paper was published in 1850s through the efforts of William Thomson (Lord Kelvin).

In order to understand the method developed by Green, we consider a general second order differential equation,

$$y'' + P(x) y' + Q(x) y = R(x), \quad (6.11.1)$$

defined over the interval $a \leq x \leq b$. If this equation has two linearly independent solutions $y_1(x)$, and $y_2(x)$, then we can find its particular integral by the method of variation of parameters as,

$$y(x) = y_1(x) \int \frac{-y_2(x) R(x)}{W(x)} dx + y_2(x) \int \frac{y_1(x) R(x)}{W(x)} dx, \quad (6.11.2)$$

where, $W(x)$ is the Wronskian of the solutions $y_1(x)$, and $y_2(x)$. We evaluate the first integral in the above equation over the interval $b \leq t \leq x$ while changing the sign and reversing the limits of integration. Moreover, we evaluate the second integral over the interval $a \leq t \leq x$. Accordingly, we obtain,

$$y(x) = y_2(x) \int_a^x \frac{y_1(t) R(t)}{W(t)} dt + y_1(x) \int_x^b \frac{y_2(t) R(t)}{W(t)} dt. \quad (6.11.3)$$

Since $y_2(x)$ is not involved in the first integral and $y_1(x)$ is not involved in the second integral, they may be taken under the respective integral signs. Thus Eqn.(6.11.3) can be written as,

$$y(x) = \int_a^x \frac{y_1(t) y_2(x)}{W(t)} R(t) dt + \int_x^b \frac{y_1(x) y_2(t)}{W(t)} R(t) dt. \quad (6.11.4)$$

This equation can be written as,

$$y(x) = \int_a^b G(x, t) R(t) dt, \quad (6.11.5)$$

where, the function $G(x, t)$ is called the **Green's function** for the differential equation (6.11.1), defined over the interval $a \leq x \leq b$. This function is defined as,

$$G(x, t) = \begin{cases} \frac{y_1(t) y_2(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1(x) y_2(t)}{W(t)}, & x < t \leq b. \end{cases} \quad (6.11.6)$$

It may be seen from the above equation that $G(x, t)$ is a continuous function of x over the interval $a \leq x \leq b$. Differentiating this equation w.r.t. x we obtain,

$$\frac{\partial}{\partial x} G(x, t) = G_x(x, t) = \begin{cases} \frac{y_1(t) y_2'(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1'(x) y_2(t)}{W(t)}, & x < t \leq b. \end{cases} \quad (6.11.7)$$

This equation indicates that as t increases across $t = x$, the function $G_x(x, t)$ is discontinuous and experiences the jump,

$$G_x(x, x_+) - G_x(x, x_-) = \frac{y_1'(x) y_2(x) - y_1(x) y_2'(x)}{W(x)} = \frac{-W(x)}{W(x)} = -1,$$

where, x_+ is the limit as t decreases to x , and x_- is the limit as t increases to x .

Now, let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the homogeneous differential equation, associated with the non-homogeneous differential equation (6.11.1), such that at $x = a$, the solution $y_1(x)$ satisfies the homogeneous boundary condition,

$$k_1 y_1(a) + k_2 y_1'(a) = 0, \quad (6.11.8)$$

and, at $x = b$, the solution $y_2(x)$ satisfies the homogeneous boundary condition,

$$k_3 y_2(b) + k_4 y_2'(b) = 0. \quad (6.11.9)$$

Then $G(x, t)$ also satisfies these same homogeneous boundary conditions. Moreover, differentiation of Eqn.(6.11.5) w.r.t. x , using Leibnitz rule, shows that the solution $y(x)$ also satisfies the homogeneous boundary conditions (6.11.8) and (6.11.9). On combining these results we get,

$$y(x) = \int_a^b G(x, t) R(t) dt, \quad \text{with } G(x, t) = \begin{cases} \frac{y_1(t) y_2(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1(x) y_2(t)}{W(t)}, & x < t \leq b, \end{cases} \quad (6.11.10)$$

is the solution of the boundary value problem for the non-homogeneous linear second-order differential equation (6.11.1).

While using the above mentioned approach, unless the Green's function is required, it is usually more convenient to obtain the solution directly from result (6.11.3). The advantage of the Green's function is that it characterizes all the essential features of the differential equation without reference to the non-homogeneous term $R(x)$, so that once it is known, Eqn.(6.11.5) solves the homogeneous boundary value problem for any function $R(x)$.

Fundamental Properties of the Green's Function over the Interval $[a, b]$

- The piece-wise defined Green's function $G(x, t)$ given by Eqn.(6.11.6) satisfies the differential equation in the respective intervals, $a \leq t \leq x$, and $x \leq t \leq b$.
- $G(x, t)$ is continuous function of x for $a \leq x \leq b$.
- $G(x, t)$ satisfies the homogeneous boundary conditions (6.11.8) and (6.11.9).
- The function $G_x(x, t)$ is continuous for $a \leq t \leq x$, and $x \leq t \leq b$, but it is discontinuous across $x = t$ where it experiences the jump,

$$G_x(x, x_+) - G_x(x, x_-) = -1.$$

Example 6.11.1

Find the Green's function for the differential equation,

$$x^2 y'' - 2xy' + 2y = 3x^2, \quad (6.11.11)$$

and use it to solve the boundary value problem subject to the conditions,

$$y(1) = 0, \quad y'(2) = 0. \quad (6.11.12)$$

Solution

The homogeneous differential equation associated with the given differential equation is,

$$x^2 y'' - 2xy' + 2y = 0. \quad (6.11.13)$$

Using the method for the solution of Cauchy-Euler equation we find that this equation has two linearly independent solutions, $y_1 = x$, and $y_2 = x^2$, so that we obtain the general solution as,

$$y = c_1 y_1 + c_2 y_2 = c_1 x + c_2 x^2. \quad (6.11.14)$$

For the solution $y_1(x)$, we must use the form of this solution that satisfies the first boundary condition, $y(1) = 0$. Thus Eqn.(6.11.14) gives $c_2 = -c_1$. Accordingly, it yields,

$$y_1(x) = c_1 (x - x^2).$$

For simplicity we may take $c_1 = 1$, so that the above solution becomes,

$$y_1(x) = x - x^2. \quad (6.11.15)$$

For the second linearly independent solution $y_2(x)$, we again use Eqn.(6.11.14) subject to the second boundary condition $y'(2) = 0$. Since $y'(x) = c_1 + 2c_2 x$, this boundary condition yields $c_1 = -4c_2$. Thus Eqn.(6.11.14) gives,

$$y_2(x) = -c_2 (4x - x^2).$$

Again, for simplicity we take $c_2 = -1$, and accordingly, obtain the second linearly independent solution of the given boundary value problem as,

$$y_2(x) = 4x - x^2. \quad (6.11.16)$$

Using solutions (6.11.15) and (6.11.16) we obtain the Wronskian $W(t)$ as,

$$W(t) = \begin{vmatrix} t-t^2 & 4t-t^2 \\ 1-2t & 4-2t \end{vmatrix} = 3t^2. \quad (6.11.17)$$

Therefore, the Green's function for the given differential equation is,

$$G(x, t) = \begin{cases} \frac{y_1(t) y_2(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1(x) y_2(t)}{W(t)}, & x < t \leq b \end{cases} = \begin{cases} \frac{(t-t^2)(4x-x^2)}{3t^2}, & 1 \leq t \leq x \\ \frac{(4t-t^2)(x-x^2)}{3t^2}, & x < t \leq 2. \end{cases} \quad (6.11.18)$$

In order to solve the given differential equation using the Green's function, we write it in the standard form: $y'' + P(x) y' + Q(x) y = R(x)$. Thus the given differential equation (6.11.11) can be written as,

$$y'' - \frac{2}{x} y' + \frac{2}{x^2} y = 3,$$

so that, $R(x) = 3$. The solution of this differential equation is given by the formula (6.11.10), i.e.,

$$y(x) = \int_a^b G(x, t) R(t) dt, \quad \text{with } G(x, t) = \begin{cases} \frac{y_1(t) y_2(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1(x) y_2(t)}{W(t)}, & x < t \leq b. \end{cases}$$

Therefore, we obtain the solution as,

$$\begin{aligned} y(x) &= \int_a^x \frac{y_1(t) y_2(x)}{W(t)} R(t) dt + \int_x^b \frac{y_1(x) y_2(t)}{W(t)} R(t) dt \\ &= y_2(x) \int_a^x \frac{y_1(t) R(t)}{W(t)} dt + y_1(x) \int_x^b \frac{y_2(t) R(t)}{W(t)} dt. \end{aligned}$$

Since $y_1(x) = x - x^2$, $y_2(x) = 4x - x^2$, and $R(x) = 3$, so the above formula gives the solution as,

$$\begin{aligned} y(x) &= (4x - x^2) \int_1^x \frac{(t - t^2)}{3t^2} 3 dt + (x - x^2) \int_x^2 \frac{(4t - t^2)}{3t^2} 3 dt \\ &= (4x - x^2) [\ln x - x + 1] + (x - x^2) [4 \ln 2 - 2 - 4 \ln x + x]. \end{aligned}$$

On simplification it gives,

$$y(x) = x^2 (\ln x^3 - \ln 16 - 2) + 2x (1 + \ln 4), \quad (6.11.19)$$

which is the solution of the given boundary value problem.

Example 6.11.2

Find the Green's function for the differential equation,

$$y'' + y = -1, \quad (6.11.20)$$

and use it to solve the boundary value problem subject to the conditions,

$$y(0) = 0, \quad y(\pi/2) = 0. \quad (6.11.21)$$

Solution

The solution of the associated homogeneous differential equation, $y'' + y = 0$, satisfying the first boundary condition $y(0) = 0$ is,

$$y_1(x) = \sin x, \quad 0 \leq x \leq t, \quad (6.11.22)$$

and its solution subject to the second boundary condition $y(\pi/2) = 0$ is,

$$y_2(x) = \cos x, \quad t \leq x \leq \pi/2. \quad (6.11.22)$$

The Wronskian of $y_1(x)$ and $y_2(x)$ is given as,

$$W(t) = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -(\sin^2 x + \cos^2 x) = -1.$$

Therefore, the Green's function for the given differential equation is,

$$G(x, t) = \begin{cases} \frac{y_1(t) y_2(x)}{W(t)}, & a \leq t \leq x \\ \frac{y_1(x) y_2(t)}{W(t)}, & x < t \leq b \end{cases} = \begin{cases} \frac{\sin t \cos x}{-1}, & 0 \leq t \leq x \\ \frac{\sin x \cos t}{-1}, & x < t \leq \frac{\pi}{2} \end{cases} \quad (6.11.23)$$

Therefore, we obtain the solution as,

$$\begin{aligned} y(x) &= y_2(x) \int_a^x \frac{y_1(t) R(t)}{W(t)} dt + y_1(x) \int_x^b \frac{y_2(t) R(t)}{W(t)} dt \\ &= \cos x \int_0^x \frac{\sin t (-1)}{-1} dt + \sin x \int_x^{\pi/2} \frac{\cos t (-1)}{-1} dt \\ &= \sin x + \cos x - 1. \end{aligned}$$

6.12 The Method of Reduction of Order

This method is attributed to **Jean d'Alembert (1717-1783)**, who is well known for his work on Fourier series, partial differential equations, mechanics, and hydrodynamics. According to this method, if one non-trivial solution y_1 of an n th order homogeneous linear differential equation with variable coefficients,

$$a_n(x) y^{(n)} + a_{n-1}(x) y^{(n-1)} + \dots + a_1(x) y' + a_0 y = 0, \quad (6.12.1)$$

is known, then the search for the remaining solutions can be narrowed down to a homogeneous linear differential equation of order $(n - 1)$, according to the following theorem:

Theorem 6.12.1

If y_1 is a solution of a homogeneous linear differential equation of order n in y , then the substitution $y = \{u(x) y_1\}$ reduces this equation to a homogeneous linear differential equation of order $(n - 1)$ in $u'(x)$.

The above theorem asserts that if y_1 is a known solution of Eqn.(6.12.1) then $\{u(x) y_1\}$ is also a solution provided that $u(x)$ is a function whose derivative satisfies the associated linear equation of order $(n - 1)$ in $u'(x)$. Since this associated linear equation is homogeneous so $u'(x) = 0$ (trivial solution) is one of the solutions of this equation. On integration it yields $u = c$, where, c is an arbitrary constant, and thus we obtain a second solution, $y_2 = u y_1$. But we see that y_1 and $u y_1$ are linearly dependent. Therefore, it does not provide us with any additional information about the solutions of Eqn.(6.12.1). However, if we are able to find a nontrivial solution for u' , then after having found u by integration, the product $y_2 = u y_1$ becomes a second solution of Eqn.(6.12.1), which is linearly independent of y_1 .

It may be seen that when a nontrivial solution u' of the associated $(n - 1)^{\text{st}}$ order equation has been found, Theorem 6.12.1 can be used again. Accordingly, further solutions of this

associated equation and hence of Eqn.(6.12.1) can be found by solving an associated equation of order $(n - 2)$. This process can be continued to find the solutions of the successive reduced equations.

Reduction of Order Method for a Second Order Equation

For $n = 2$, Eqn.(6.12.1) becomes,

$$a_2(x) y'' + a_1(x) y' + a_0 y = 0.$$

Dividing this equation by $a_2(x)$, we can write it in the standard form,

$$y'' + P(x) y' + Q(x) y = 0, \quad (6.12.2)$$

where, $P(x)$ and $Q(x)$ are continuous on some interval I . Suppose $y_1(x)$ is a known solution of this equation on an interval I , and that $y_1(x) \neq 0$ for all x in this interval. To find a general solution of this second order differential equation, we must have two linearly independent solutions. In order to find the second linearly independent solution, we attempt to find a solution of the form, $y = u(x) y_1$, and solve for $u(x)$. Differentiating it we obtain,

$$y' = u' y_1 + u y_1', \quad \text{and} \quad y'' = u'' y_1 + 2 u' y_1' + u y_1''.$$

Substituting these values in Eqn.(6.12.2) we obtain,

$$\begin{aligned} y'' + P(x) y' + Q(x) y &= (u'' y_1 + 2 u' y_1' + u y_1'') + P(x) \{ u' y_1 + u y_1' \} + Q(x) y \\ &= u \{ y_1'' + P(x) y_1' + Q(x) y_1 \} + y_1 u'' + \{ 2 y_1' + P(x) y_1 \} u' = 0. \end{aligned}$$

But $y_1'' + P(x) y_1' + Q(x) y_1 = 0$, as y_1 is a given solution of Eqn.(6.12.2). Thus the above equation reduces to,

$$y_1 u'' + \{ 2 y_1' + P(x) y_1 \} u' = 0.$$

If we let $w = u'$, then this equation becomes a first-order linear differential equation in w ,

$$y_1 w' + \{ 2 y_1' + P(x) y_1 \} w = 0. \quad (6.12.3)$$

Separating the variables it gives,

$$\frac{dw}{w} = - \left(2 \frac{y_1'}{y_1} + P \right) dx.$$

Integrating it we get,

$$\ln w = - \ln y_1^2 - \int P(x) dx + \ln c_1,$$

or,

$$w y_1^2 = c_1 e^{-\int P dx}, \quad \text{or,} \quad w = c_1 \frac{e^{-\int P dx}}{y_1^2}.$$

But $w = u'$. Therefore, on integration the above equation gives,

$$u = c_1 \int \frac{e^{-\int P dx}}{y_1^2} dx + c_2.$$

For simplicity we choose $c_1 = 1$, and $c_2 = 0$. Therefore, using $y = u(x) y_1$, we find the second solution of Eqn.(6.12.2) as,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx. \quad (6.12.4)$$

It may be verified that the function y_2 , given above satisfies Eqn.(6.12.2), and that y_1 and y_2 are two linearly independent solutions on any interval I on which y_1 is not zero.

Example 6.12.1

Find a second linearly independent solution of the differential equation,

$$y'' + 6y' + 9y = 0, \quad (6.12.5)$$

given that $y = e^{-3x}$ is a solution of this equation.

Solution

Comparing Eqn.(6.12.5) with the standard second-order linear differential equation (6.12.2) we see that $P(x) = 6$. Since $y_1(x) = e^{-3x}$ is a known solution of the given differential equation, so using the formula (6.12.4) we obtain,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{-3x} \int \frac{e^{-\int 6 dx}}{\{e^{-3x}\}^2} dx = e^{-3x} \int \frac{e^{-6x}}{e^{-6x}} dx = x e^{-3x},$$

which is the second linearly independent solution of the given differential equation.

Example 6.12.2

Given that $y_1 = x$ is a solution of the differential equation,

$$x^3 y''' - 3x^2 y'' + x(6 - x^2) y' - (6 - x^2) y = 0. \quad (6.12.6)$$

Show $y = ux$ will also be a solution of this equation provided that u' satisfies the equation $(u')'' - u' = 0$.

Solution

From the given solution, $y = ux$, we obtain on differentiation,

$$y' = x u' + u, \quad y'' = x u'' + 2 u', \quad \text{and,} \quad y''' = x u''' + 3 u''.$$

Substituting these values in the given differential equation (6.12.6), we see that $y = ux$, will also be a solution of this equation if,

$$x^3 (x u''' + 3 u'') - 3x^2 (x u'' + 2 u') + x(6 - x^2) (x u' + u) - (6 - x^2) x u = 0.$$

On simplification it gives,

$$x^4 u''' - x^4 u' = 0.$$

Since $x \neq 0$, so $u''' - u' = 0$, i.e.,

$$(u')'' - u' = 0, \quad (6.12.7)$$

which is what is required.

In order to find the remaining solutions of the given differential equation we can solve (6.12.7) for u' . Since its auxiliary equation has the roots 1, -1, so it has the solutions $u' = e^x$, and $u' = e^{-x}$. Integrating them we get,

$$u_1 = e^x, \quad \text{and} \quad u_2 = -e^{-x}.$$

Therefore, using $y = u x$, we obtain two more solutions of the given differential equation as,

$$y_2 = x e^x, \quad \text{and} \quad y_3 = -x e^{-x}.$$

Example 6.12.3

Given that $y_1 = x^{-1/2}$ is a solution of the differential equation,

$$4x^2 y'' + 8x y' + y = 0, \quad x > 0, \quad (6.12.8)$$

Find the second linearly independent solution of this equation.

Solution

In order to reduce the given differential equation to the standard form (6.12.2), we divide it by $4x^2$ to get,

$$y'' + \frac{2}{x} y' + \frac{1}{4x^2} y = 0.$$

On comparing this equation with the standard form (6.12.2) we see that $P(x) = \frac{2}{x}$. Therefore, for $y_1 = x^{-1/2}$, the formula (6.12.4) gives the second solution of the given differential equation as,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = x^{-1/2} \int \frac{e^{-\int (2/x) dx}}{\{x^{-1/2}\}^2} dx = x^{-1/2} \int \frac{e^{-2 \ln x}}{x^{-1}} dx = x^{-1/2} \int \frac{1}{x} dx,$$

i.e.,

$$y_2 = x^{-1/2} \ln x, \quad x > 0.$$

EXERCISE – 6.5

[The Methods of Variation of Parameters, Green's Function, and Reduction of Order]

Note. Answers are given against the respective questions within square brackets.

Solve the following differential equations by the method of *variation of parameters*:

1. $y'' = 1$; $[y = c_1 + c_2 x + x^2/2]$
2. $y'' + y = 1$; $[y = c_1 \cos x + c_2 \sin x + 1]$
3. $y'' - y = x + 1$; $[y = c_1 e^x + c_2 e^{-x} - x - 1]$
4. $y'' - y = 5e^x$; $[y = c_1 e^x + c_2 e^{-x} + 5xe^x/2]$
5. $y'' + y' = 4x$; $[y = c_1 + c_2 e^{-x} + 2x^2 - 4x]$
6. $y'' + y' = e^{-x}$; $[y = c_1 + c_2 e^{-x} - xe^{-x}]$
7. $y'' + y = \sin x$; $[y = A \cos x + B \sin x - (x \cos x)/2]$
8. $y'' - 2y' + 2y = e^x \sin x$; $[y = e^x \{A \cos x + B \sin x - (x \cos x)/2\}]$
9. $y'' + y = \sec x \tan x$; $[y = A \cos x + B \sin x - x \cos x + \sin x \ln |\sec x|]$
10. $y'' + y = \operatorname{cosec} x$; $[y = A \cos x + B \sin x + x \cos x + \sin x \ln |\sin x|]$
11. $y'' - 3y' + 2y = 1/(1 + e^{-x})$; $[y = c_1 e^{2x} + c_2 e^x + (e^x + e^{2x}) \ln(1 + e^{-x})]$
12. $y'' + 2y' + y = e^{-x} \ln x$; $[y = c_1 e^{-x} + c_2 xe^{-x} + (x^2 e^{-x} \ln |x|)/2 - (3x^2 e^{-x})/4]$
13. $y'' + y' - 2y = x e^x$; $[y = c_1 e^x + c_2 e^{-2x} + e^x/27 - xe^x/9 + x^2 e^x/6]$
14. $y'' - y' - 2y = e^{2x}$; $[y = c_1 e^{2x} + c_2 e^{-x} + (xe^{2x})/3]$
15. $y'' + 5y' + 6y = x^2 e^{-2x}$; $[y = c_1 e^{-2x} + c_2 e^{-3x} - 2e^{-2x} + 2xe^{-2x} - x^2 e^{-2x} + x^3 e^{-2x}/3]$
16. $y'' + 4y = \tan 2x$; $[y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} \cos 2x \ln |\sec 2x + \tan 2x|]$
17. $xy'' - 2xy' + xy = 2e^x$ $[y = c_1 e^x + c_2 xe^x - 2xe^x + 2xe^x \ln |x|]$
18. $y'' + 4y' + 5y = xe^{-2x} \cos x$; $[y = c_1 e^{-2x} \cos x + c_2 e^{-2x} \sin x + (xe^{-2x} \cos x)/4 + (x^2 e^{-2x} \sin x)/4]$
19. $y'' + 16y = x^2 e^x$; $[y = c_1 \cos 4x + c_2 \sin 4x - (26 e^x)/4913 - (4 xe^x)/289 + (x^2 e^x)/17]$
20. $y'' + 3y' + 2y = 3/(1 + e^x)$; $[y = c_1 e^{-2x} + c_2 e^{-x} + 3e^{-2x} \ln(1 + e^x) + 3e^{-x} \ln(1 + e^x)]$
21. $y'' + y = \sec^2 x$; $[y = c_1 \cos x + c_2 \sin x - 1 - \cos x + 2 \tanh^{-1}\{\sin x/(1 + \cos x)\} \sin x]$
22. $y''' - 3y' + 2y = 2x^2 + 3x$; $[y = e^{-x}(A \cos x + B \sin x) + x^2/2 - x + 1]$
23. $y''' - y'' + y' - y = 4 \sin x$; $[y = c_1 e^x + c_2 \sin x + c_3 \cos x - x \sin x + x \cos x]$

In the following problems verify that $y_1(x)$ and $y_2(x)$ are linearly independent solutions of the homogeneous form of the given differential equation. Use these solutions to find a particular integral (**Note: The coefficient of y'' must always be 1 for which preliminary division may be required** in some of the problems):

24. $x^2 y'' - 2xy' + 2y = x^3 \sin x$, $y_1(x) = x$, $y_2(x) = x^2$; $[y_p = -x \sin x]$
25. $x^2 y'' + 3xy' - 3y = \sqrt{x}$, where $y_1(x) = x$ and $y_2(x) = x^{-3}$; $[y_p = -(4\sqrt{x})/7]$
26. $x^2 y'' + xy' - 4y = x(x + x^3)$, $y_1(x) = x^2$, $y_2(x) = x^{-2}$; $[y_p = (x^2 \ln x)/4 + x^4/12]$
27. $(1 - x^2)y'' - xy' + 4y = x$, where $y_1(x) = 2x^2 - 1$, and $y_2(x) = x\sqrt{x^2 - 1}$; $[y_p = x/3]$
28. $(1 - x)y'' + xy' - y = 2(x - 1)e^{-x}$, where $y_1(x) = x$, and $y_2(x) = e^x$; $[e^{-x}(1/2 - x)]$
29. $y'' + y'/x + (1 - 1/4x^2)y = 1/\sqrt{x}$, $y_1(x) = x^{-1/2} \sin x$, $y_2(x) = x^{-1/2} \cos x$; $[y_p x^{1/2}]$
30. $y'' - y = e^{-x^2}$, $y_1(x) = e^x$, $y_2(x) = e^{-x}$; $[y_p = \int \sinh(x - t) e^{-t^2} dt]$

Find the Green's functions for the following differential equations, subject to the associated boundary conditions.

31. $y'' = R(x)$, $y(0) = 0$, and $y(1) = 0$ $[y_1(x) = x, y_2(x) = 1 - x, W(t) = -1,$
 $G(x, t) = \begin{cases} t(x-1), & 0 \leq t \leq x \\ x(t-1), & x < t \leq 1 \end{cases}]$
32. $y'' = R(x)$, $y(0) = 0$, and $y'(1) = 0$
33. $y'' + \lambda^2 y = R(x)$, $y(0) = 0$, and $y(1) = 0$ $[y_1(x) = \sin \lambda x, y_2(x) = \frac{\sin \lambda(1-x)}{\sin \lambda}, W(t) = -\lambda,$
 $G(x, t) = \begin{cases} \frac{\sin \lambda t \sin \lambda(x-1)}{\lambda \sin \lambda}, & 0 \leq t \leq x \\ \frac{\sin \lambda x \sin \lambda(t-1)}{\lambda \sin \lambda}, & x < t \leq 1. \end{cases}]$
34. $y'' + \lambda^2 y = R(x)$, $y(0) = 0$, and $y'(1) = 0$

Solve the following differential equations by means of Green's functions,

35. $x^2 y'' + x y' - y = x^2 e^{-x}$, $y(1) = 0$, and $y(2) = 0$; $[y_1(x) = 1 - \frac{1}{x}, y_2(x) = x - \frac{4}{x}, W(t) = 6/t$
 $y(x) = \frac{e^{-x}}{x}(1+x) + \frac{1}{xe^2}(1-x^2) + \frac{2}{3xe}(x^2-4)]$

36. $x^2 y'' + 2x y' - 2y = x^3$, $y(1) = 0$, and $y(2) = 0$

37. $x^2 y'' - 3xy' + 3y = x^2 \ln x$, $y'(1) = 0$, and $y(2) = 0$ [$y_1(x) = 3x - x^3$, $y_2(x) = 4x - x^3$, $W(t) = 2t^3$
 $y(x) = x(4 - x \ln x - x^2) - 2x \ln 2(3 - x^2)$]

38. $x^2 y'' - 3x y' = x^2$, $y(1) = 0$, and $y(2) = 0$

Use the Reduction of Order Method with the solution $y_1(x)$ to find the second linearly independent solution for the following differential equations:

39. $y'' - 5y' + 6y = 0$, $y_1(x) = e^{3x}$. [$y_2(x) = e^{2x}$]

40. $y'' - 4y' + 4y = 0$, $y_1(x) = e^{2x}$ [$y_2(x) = x e^{2x}$]

41. $y'' + 4y = 0$, $y_1(x) = \cos 2x$ [$y_2(x) = \sin 2x$]

42. $x^2 y'' + 4x y' - 4y = 0$, $y_1(x) = x^{-4}$ [$y_2(x) = x$]

43. $x^2 y'' + 3x y' + y = 0$, $y_1(x) = x$ [$y_2(x) = \frac{1}{x} \ln x$]

44. $y'' + 4y' + 5y = 0$, $y_1(x) = e^{-x} \cos x$ [$y_2(x) = e^{-x} \sin x$]

45. $x y'' + 2y' + xy = 0$, $y_1(x) = \frac{\sin x}{x}$ [$y_2(x) = \frac{\cos x}{x}$]

46. Suppose that $f(x)$ is a solution of the differential equation: $y'' + P(x) y' + Q(x) y = 0$. Show that $f(x)$ and the solution: $f(x) \int \frac{e^{-\int P dx}}{\{f(x)\}^2} dx$, obtained by reduction of order are linearly independent. (Hint: Use the Wronskian).